

Babul Kumar

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SKILLS

Languages: Python, C++, JavaScript, TypeScript, Java
Frameworks: React.js, Tailwind CSS, Streamlit
AI/ML: Machine Learning, NLP, Generative AI, Gemini API, Scikit-learn
Tools & Platforms: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Chrome Extension (Manifest V3)
Soft Skills: Problem-Solving, Team Collaboration, Adaptability, Leadership

PROJECTS

Flight Delay Prediction | [Live Demo](#) | [Github](#) May 2026
Engineered an ML pipeline applying OOF target encoding and temporal, geospatial, and operational feature engineering to flight delay data.
Evaluated Logistic Regression, Polynomial Regression, and AdaBoost models using cross-validation, F1-score, and ROC-AUC to select the highest-performing classifier.
Deployed a Streamlit web application for real-time flight delay prediction and risk assessment.
Tech: Python, Pandas, NumPy, Scikit-learn, Streamlit

FloatTube AI | [Github](#) Aug 2026
Built a Manifest V3 Chrome extension featuring Picture-in-Picture, transcript search, timestamped notes, bookmarks, and a distraction-free Focus Mode.
Integrated Google Gemini to generate transcript-grounded summaries, Q&A, flashcards, and automated quizzes from video content.
Architected a modular React + TypeScript codebase supporting multiple video platforms.
Tech: TypeScript, React, Gemini API, IndexedDB, Chrome MV3, Vite

AI Product Review Analyzer | [Github](#) Aug 2026
Developed a Gemini-powered analysis system generating review summaries, sentiment analysis, pros/cons, and structured insights.
Implemented reliable LLM processing with JSON schema validation, retry handling, defensive parsing, and API-key rotation to ensure production stability.
Designed a responsive React interface using Tailwind CSS and Framer Motion for an interactive user experience.
Tech: TypeScript, React, Gemini API, Vite, Tailwind CSS

TRAINING

Cybersecurity Training | CipherSchools | [Certificate](#) | [Github](#) Jun 2026 – Aug 2026
Studied core cybersecurity concepts including common security threats, vulnerabilities, network security, and defensive techniques.
Gained hands-on exposure to ethical hacking and cybersecurity tools for identifying and analyzing security vulnerabilities.
Final Project: Developed a keylogger application to study keystroke monitoring, and defensive countermeasures.

CERTIFICATIONS

Database Management System Part 1 | Infosys | [Certificate](#) Aug 2026
Data Structures & Algorithms with C++ | Coding Tantra | [Certificate](#) Jan 2025
React.js | Coding Tantra | [Certificate](#) Mar 2025

EXTRA-CURRICULAR ACTIVITIES

Competed in Hack the Vibe 2025, a hackathon conducted by THC. [Certificate](#)
Represented Lovely Professional University at the Serve-Smart Hackathon at Jagriti 2026 by IIT (BHU) Varanasi. [Certificate](#)
Engaged in Kaggle competitions, applying machine learning, data preprocessing, feature engineering, and model evaluation techniques. [Kaggle Profile](#)

EDUCATION

Lovely Professional University <i>Bachelor of Technology — Computer Science and Engineering CGPA: 7.95</i>	Phagwara, Punjab 2024 – Present
Mount Carmel International School <i>Intermediate Percentage: 63.3%</i>	Samastipur, Bihar 2023 – 2024
Central Public School <i>Matriculation Percentage: 87.6%</i>	Samastipur, Bihar 2021 – 2022